CLASS 11-CBSE MATHEMATICS

CHAPTER: PERMUTATIONS AND COMBINATIONS-SET-1

TIME : 1Hr **Mx.Marks:30**

SECTION A

Q.I Answer the following questions **[1X3 = 3 MARKS]**

1. Find the number of permutations of the letters of the word ALLAHABAD.
2. 20 persons including Mr. A and Mrs. A are invited to a party. In how many ways the host, the hostess and guests can be arranged around a circular table such that the host & the hostess sit together flanked by Mr. A and Mrs. A.
3. Compute $\frac{8!6!}{2!}$

SECTION B

Q.II Answer the following questions **[2X5 = 10 MARKS]**

4. In how many ways the 52 cards of a pack can be distributed
   (a) Equally among 4 heaps
   (b) Equally among 4 players
   (c) In 4 sets, three of 17 cards each & fourth having 1 card only?
5. Out of 6 boys and 4 girls a group of 7 is to be formed. How many such groups are possible if boys are to be in majority?
6. Calculate the number of diagonals that can be formed by a polygon of 50 sides.
7. Find r, if $5\,{}^{4}Pr = 6\,{}^{5}P(r-1)$
8. There are p intermediate railway stations on a railway line from one terminal to another. In how many ways can a train stop at three of these intermediate stations if no two of these stations (where it stops) are to be consecutive?

SECTION C

Q.III Answer the following questions **[3X3 = 9 MARKS]**

9. A bag contains 5 black and 6 red balls determine the number of ways in which 2 black and 3 red balls can be selected.
10. Five balls of different colors are to be placed in three boxes of different sizes. Each box can hold all the 5 balls. In how many different ways can we place the balls so that no box remains empty?
11. The number of 4 - digit numbers formed with the digits 1,2,3,4,5,6,7 which are divisible by

25 is (Rept. not allowed).

SECTION D

Q.IV Answer the following questions **[4X2 = 8 MARKS]**

12. Given five different green dyes four different blue dyes and three different red dyes how many combinations of dyes can be chosen taking at least one green and one blue dyes.
13. A group consists of 4 girls and 7 boys. In how many ways can a team of 5 members be selected if the team has:
    (i) no girl? (ii) at least one boy and one girl? (iii) at least 3 girls?